

Virtual Control Interface: Discover and Control IoT Devices Intuitively Through AR Glasses with Multi-model Interactions

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ABSTRACT

The number of smart home devices will increase exponentially. The current Internet of Things (IoT) control interfaces on smartphones are spatially separated from the devices they operate, making them less intuitive and progressively more complicated. We developed VCI, a Virtual Reality (VR) simulation for HCI researchers to explore multimodal interactions with IoT in a future smart home setting using virtual control interfaces projected on emulated AR glasses.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality Human-centered computing—Human computer interaction (HCI)—Interactive systems and tools;

1 INTRODUCTION

In 2019, U.S. households owned an average of 11 connected devices [2]. This number will only go up as people get more accustomed to the concept of a connected future.

Consumers of Internet of Things (IoT) devices currently control those through standard user interfaces such as apps on smartphones, control hubs on walls, and AI virtual assistants via smart speakers. These are broadly accessible and offer remote operation compared to traditional non-IoT devices (e.g., physical buttons, knobs, screens). However, they pose some challenges for the users as they are spatially separated from the devices they operate. Moreover, as users add more devices to the network, these interfaces become less intuitive and more complicated since they need to keep track of the associations between icons, custom names, and physical devices.

Future trends suggest AR glasses could become the next consumer terminal. If AR glasses were to replace smartphones in the near future, we would need to develop new AR interfaces to control IoT devices.

Developing new AR-IoT interfaces can be quite challenging as it takes a lot of effort, time, and resources to study complex HCI problems related to AR and IoT in real life. Setting up IoT devices with desired functionalities, establishing connections and communication between AR and IoT devices requires many engineering hours, complex infrastructures, and big budgets. Moreover, current AR devices are hindered by limited field of view (FOV) processing power and run time. However, within a VR simulation, these restrictions are lifted.

We have developed a VR based simulation framework called Virtual Control Interface (VCI) to offer the HCI community the ability to perform speculative research and explore complex UX/UI and multi-modal interactions with AR and IoT devices on everyday scenarios that can take place in a simulated future smart home.

VCI incorporates hand gestures, eye tracking (gaze control), and voice recognition as input to control smart home devices and provide visual feedback projected on emulated AR glasses.

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2 IOT DEVICE LOCALIZATION

One of the advantages of VCI is the ability to spatially associate control interfaces with the smart devices they control to reduce the complexity of discovering and using smart devices.

AR glasses need to know each device's precise location to make the association and realize this advantage. In VR simulation, we can mimic localization API by adding a "tag" to each smart device object and obtain each device's location directly through the Unity engine. If researchers later want to test simulated AR-IoT interactions in real life, they could leverage the Ultra-wideband (UWB) technology to obtain smart devices' relative positioning. UWB is a localization technology that can achieve an accuracy of 10-30 centimeters [1]. Apple adopted it in the iPhone 11 series, followed by Chinese smartphone companies such as Xiaomi, proving the technology's value in the consumer market. However, almost none of the available IoT devices has adopted UWB yet.

3 INPUT METHODS

There are currently three types of user inputs available to interact with VCI: hand gestures, eye tracking, and voice commands. Hand tracking and voice commands are readily available in standalone VR headsets such as Oculus Quest. To mimic eye tracking, we put a grey dot at the center of the screen as the line of sight. Users can turn their heads to move this grey dot.

In the following sections, we will discuss how we use these inputs to explore multimodal interaction scenarios.

4 DISCOVER SMART HOME DEVICES WITH INDICATOR POINTS

VCI uses indicator points to inform a user where smart devices are located. These are grey-boarded white dots containing an icon at the center, as displayed in our demo. The icon indicates what type of device it is controlling, so the user is confident they pick the device they want to control. The users can discover smart home devices by looking around and finding the indicator points associated with them.

Imagine an invisible sphere, centered at the user's location with a radius of 5 meters. VCI places an indicator point at the intersection between the sphere's surface and the line between the user and the smart device.

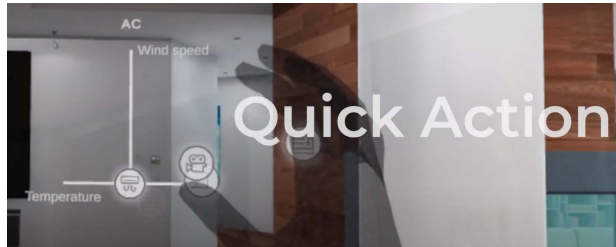
Following an initial heuristic evaluation, we kept indicator points at the same distance from the user to prevent those attached to devices far from the user to become extremely small and unusable.

Indicator points are the starting point for most interactions a user will perform with VCI through Quick Actions, Smart Voice Commands, and Full Control Panels.

5 QUICK ACTIONS WITH GESTURE RECOGNITION

5.1 Overview

Quick actions allow users to control primary settings on smart devices rapidly. The users do not need to look at the devices to perform such interactions. The user can move their hand towards a device, pinch the thumb, index finger together, and drag horizontally or vertically to change specific settings. Examples include color and



brightness of light, temperature and fan speed of AC, and sunblind height.

5.2 Attaching Indicator Points to Hands

Using hands as pointers is not as precise as using a mouse. Rather than requiring the user to perform precise hand movements to pick an indicator point, VCI attaches the interface to the user's hand based on average proximity from a target. When the angle of the line between the user's head and hand is within a certain degree between the user's head and a smart device, the smart device's indicator point attaches automatically to the user's hand. If multiple smart devices are close together, VCI will pick the one with the smallest approximate angle.

5.3 Performing Controls

Quick action currently supports two controls by dragging the indicator point horizontally or vertically. For the vertical axis, dragging down reduces the corresponding value and vice versa. For the horizontal axis, dragging left reduces the corresponding value and vice versa. Users can easily change values by sliding the point in different directions. A text label shows an aspect of the smart device the user can control along that axis—examples including temperature for AC, brightness for light, height for blinds.

It is impossible to move a hand only along one axis; there will always be slight maneuvers on the other axis. Therefore, VCI predicts the user's intention by analyzing the user's hand movement and lock the axis that users intend to control.

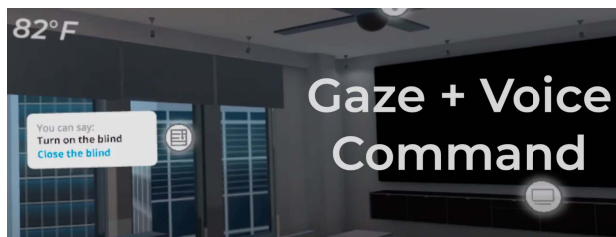
We are currently developing a third gestural control that will allow users to spread their fingers out and rotate the palm.

5.4 Feedback

The system provides some immediate visual feedback. When the user selects an indicator point by pinching, the indicator point gradually becomes smaller; the icon fades out; the color of the point turns blue with a click sound played.

When the user drags an indicator point along an axis, VCI shows the new value on top of the indicator point.

6 SMART VOICE COMMANDS WITH GAZE CONTROL



6.1 Overview

In case the user prefers not to use their hands, they can still choose to use VCI's smart voice command system. By looking at a smart home device, VCI knows which device a user wants to control and provides personalized suggestions for that device's possible commands.

6.2 How Eye Tracking differentiates VCI from Generic Voice Assistants

Two problems that arise when using generic voice assistants are ambiguity and complexity.

If there are ten blinds in the living room, how can the user remember which one is the "blind 1" and control just that blind? If a user wants to turn up the range hood's fan speed, how can the voice assistant know the user's intention is not to turn the fan speed up for the AC?

It is possible to remember the name and commands for a dozen smart devices, but what if there are hundreds of them?

With VCI's smart voice command with gaze control, VCI will acknowledge the user's intention to control to reduce ambiguity and provide suggestions to remove the complexity of using voice assistants.

6.3 Controls

Unlike quick actions, the user needs to look at an indicator point to use smart voice control. VCI tracks the user's eye direction and captures the furniture that the user is looking at. A panel shows up next to the indicator point, including a few possible voice commands the user might need. Then users can say what they want to do with the device.

6.4 Feedback

When a voice command is recognized, the corresponding prompt will change to a blue color with a sound effect played.

7 ACHIEVING OTHER FUNCTIONS WITH FULL CONTROL PANEL

VCI does not aim to demonstrate an interface that can handle all possible operations required to perform on IoT devices. It is also not possible with quick actions, and smart voice commands alone. (Probably reverse the tone to show what it can do?) VCI does include a few other quarks in the demo, but they will not be enough to encapsulate the complex nature of a universal interface.

To solve this problem, VCI provides a full control panel where all other functions can reside. The user can access the full control panel by dragging an indicator point towards him. IoT device manufacturers can customize full control panels for each product to provide a full set of functionality to their customers.

8 EVALUATION

We found out people with no prior experience using VR or AR could utilize VCI within a few minutes through testing.

We did find problems that needed future work. When multiple devices are close together, their indicator points overlap, making it hard to pick which one to operate. We will implement grouping or other logic to tackle this issue. We also need to work on controlling devices out of sight.

Other than resolving problems, we plan to explore more scenarios such as user authentication, personalized user interface for different user groups, interaction data processing, and visualization in the future.

REFERENCES

- [1] Ultra-wideband technology.
- [2] T. Spangler. U.s. households have an average of 11 connected devices - and 5g should push that even higher, Dec 2019.